

Test Results, Observations and Use Case Summary – Privilege Escalation Detection

1. Use Case Objective

The objective of this use case was to detect privileged group membership modifications within a Windows Active Directory environment and automatically generate enriched incidents within Splunk SOAR.

The solution focused on monitoring additions to:

- Domain Admins
- Local Administrators

using Windows Security Events and automated response workflows.

2. Test Environment

SIEM Components

- Splunk Enterprise Search Head
- Splunk Enterprise Indexer
- Splunk Heavy Forwarder

SOAR Components

- Splunk SOAR 7.1.0.225

Data Sources

- Windows Server 2022 Domain Controller
- Windows 11 Client

Cloud Platform

- AWS EC2

3. Test Scenario

The following privilege escalation scenarios were executed.

Scenario 1 – Domain Privilege Escalation

A domain user account was added to the Domain Admins group.

Expected Result:

- Event ID 4728 generated.
- Splunk alert triggered.
- Event forwarded to SOAR.
- Playbook executed successfully.

Scenario 2 – Local Privilege Escalation

A user account was added to the local Administrators group.

Expected Result:

- Event ID 4732 generated.
- Splunk alert triggered.
- Event forwarded to SOAR.
- Playbook executed successfully.

4. Test Results

Test Step	Expected Result	Actual Result	Status
Domain Admin membership change	Event ID 4728 generated	Successful	Pass
Local Administrator membership change	Event ID 4732 generated	Successful	Pass
Log collection	Events visible in Splunk	Successful	Pass
Field extraction	Required fields extracted	Successful	Pass
Detection logic	Alert generated	Successful	Pass
Event forwarding	Alert sent to SOAR	Successful	Pass
Playbook execution	Automated workflow executed	Successful	Pass
Note creation	Incident note added	Successful	Pass
Tag assignment	Tags added	Successful	Pass
Severity assignment	High severity applied	Successful	Pass

5. Key Observations

Observation 1

Windows Event IDs 4728 and 4732 provide reliable visibility into privileged group membership modifications.

Observation 2

Normalization of event fields improves detection consistency across different systems.

Observation 3

Splunk SOAR successfully enriched incidents and reduced manual analyst effort.

Observation 4

Automated tagging and severity assignment improve SOC triage efficiency.

6. Security Benefits

- Detection of privilege escalation activity.
- Increased visibility into administrative access changes.
- Automated incident enrichment.
- Improved SOC workflow efficiency.
- Reduced investigation time.

7. Potential False Positives and Legitimate Administrative Activities

While the detection logic successfully identifies privileged group membership modifications, not every detected event represents malicious activity.

The following scenarios may generate legitimate alerts:

Scenario 1 – Authorized User Provisioning

A system administrator may add a new employee to the Domain Admins or Administrators group as part of approved onboarding procedures.

Expected Action:

Verify change request, approval records, and administrator activity before escalating.

Scenario 2 – Temporary Privileged Access

Administrators may grant temporary elevated privileges to users for maintenance, troubleshooting, or project-related tasks.

Expected Action:

Confirm whether the privilege assignment was approved and time-bound.

Scenario 3 – Security Team Activities

Security personnel may intentionally add accounts to privileged groups during incident response, forensic investigations, or testing activities.

Expected Action:

Validate the activity against security operations records.

Scenario 4 – System Migration or Application Deployment

Automated deployment tools, service accounts, or migration activities may perform legitimate group membership modifications.

Expected Action:

Verify whether the change was associated with planned maintenance activities.

Scenario 5 – Lab and Training Environments

In testing or educational environments, administrators may intentionally perform privilege escalation activities to validate security controls.

Expected Action:

Confirm whether the activity originated from an approved testing exercise.

8. Analyst Investigation Guidance

Before classifying a privilege escalation alert as malicious, analysts should review:

- User account involved.
- Account performing the modification.
- Change management records.
- Approved maintenance windows.
- Recent help desk requests.
- Historical behavior of the affected account.
- Associated authentication and logon events.

A privilege escalation event should be considered suspicious when there is no legitimate business justification, approval record, or administrative explanation for the privileged access change.

9. Challenges Encountered

- Event field extraction differences between Domain Controller and Client logs.
- Correlation of user information across different event types.
- Event forwarding field mapping validation.
- SOAR playbook data-path configuration and testing.

10. Conclusion

This use case successfully demonstrated end-to-end privilege escalation monitoring using Splunk Enterprise and Splunk SOAR.

The implementation detected privileged group membership changes, generated security alerts, forwarded incidents to Splunk SOAR, and automatically enriched incidents with notes, tags, and severity information.

The solution provides a practical demonstration of SIEM detection engineering, security monitoring, incident enrichment, and SOC automation workflows commonly used in enterprise environments.